BIOcean5D — coral microbiome responses to reciprocal transplantation and acute heat shock

Results summary, 26 October 2026 · *Pocillopora meandrina* (POC) and *Porites rus* (POR) · Opunohu Bay (turbid, impacted) versus Tiahura (pristine), Mo'orea · All values from the knitted run of Analysis/05-AAC-CommunityAnalyses.Rmd (26 Oct 2026).

Design and dataset

Fragments of both hosts were reciprocally transplanted between the two bays and sampled on a three-point timeline — mother-colony baseline (*Genotype*), four months in situ (*4-Month-Pre-HS*), and immediately after an acute heat challenge (*Post-HS*) — alongside environmental references from free-living seawater (0.2 µm), particle-bound seawater (3 µm) and benthic sediment. After removal of low-depth samples and non-asymptotic rarefaction curves, **111 of 122 samples (91.0%) were retained**; libraries were scaled to the within-group minimum, 3,072 reads for *Pocillopora* and 2,066 for *Porites*. Community structure was tested by PERMANOVA on four distance metrics (Sørensen, Bray–Curtis, unweighted and weighted UniFrac) with permutations blocked by mother colony.

Main results

1. Compartment, not site, dominates. Host/medium (*Subtype*) explains 10.8% (Sørensen) to 23.6% (weighted UniFrac) of variation (all p = 0.001); site explains 0.95–1.55%. Site is detectable in three of four metrics (p = 0.002–0.013) but **not** in unweighted UniFrac (p = 0.105): the pool of lineages present is shared between bays, and site differences are carried by relative abundances. A significant *Subtype* × *Site* interaction in every metric (4.3–4.8%, p ≤ 0.015) justifies analysing hosts separately.

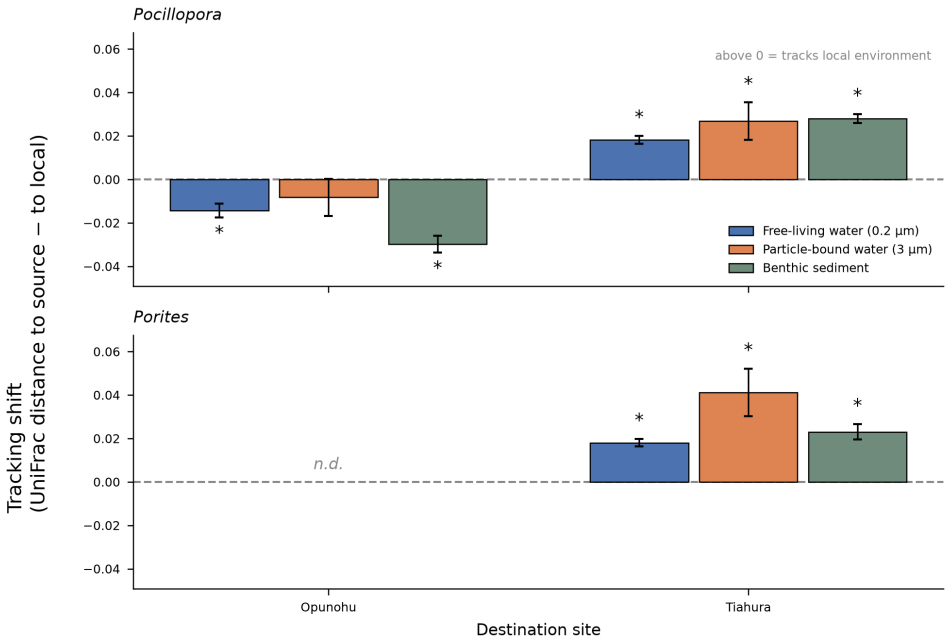
2. Site sensitivity is host-specific. With permutations blocked by colony, *Pocillopora* differs by site in three metrics — Bray–Curtis R^2 = 3.2% (p = 0.001), weighted UniFrac R^2 = 3.8% (p = 0.008), Sørensen R^2 = 1.9% (p = 0.014) — but not unweighted UniFrac (p = 0.205). *Porites* shows **no site effect in any metric** (p = 0.066–0.746).

Term	Pocillopora R^2 (p)	Porites R^2 (p)
Site	1.9–3.8% (0.001–0.205)	2.1–4.0% (0.066–0.746)
Treatment (timeline)	6.6–8.1% (0.001–0.003)	10.2–19.4% (all 0.001)
Site × Treatment	2.7–4.1% (0.018–0.788)	6.2–8.0% (0.153–0.508)

Ranges span the four distance metrics; weighted UniFrac gives the largest timeline effect in both hosts.

3. Time in the field outweighs site. The timeline term is significant in all four metrics in both hosts and is roughly twice as large in *Porites*. Post-hoc tests place the shift in the four-month interval, not the heat challenge: in *Pocillopora*, *Genotype* vs *Post-HS* is significant at both sites (p = 0.001–0.021) while *4-Month-Pre-HS* vs *Post-HS* is significant in none (p = 0.056–0.245); in Opunohu *Porites* only *Genotype* vs *4-Month-Pre-HS* survives correction (p = 0.041–0.044).

4. Fragments track local sources in Tiahura but resist in Opunohu. Comparing each out-transplanted fragment's unweighted UniFrac distance to local versus source-site references (paired Wilcoxon, FDR-corrected), every fragment moved *into* Tiahura converged on local sources (+0.018 to +0.041, FDR ≤ 0.008 in all six host × medium combinations). *Pocillopora* moved into Opunohu did the opposite, staying closer to its Tiahura source signature for free-living water (–0.014, FDR = 0.004) and sediment (–0.030, FDR = 0.0007); the particle-bound fraction was not significant (–0.008, FDR = 0.34). This asymmetry is the most distinctive result in the dataset.



Directional environmental tracking of transplanted fragments. Mean ± SE of the per-fragment difference in unweighted UniFrac distance (distance to source-site reference minus distance to local reference); values above zero indicate convergence on the destination bay's community. Asterisks mark FDR-corrected paired Wilcoxon p < 0.05. n = 13 and 15 fragments for *Pocillopora* in Opunohu and Tiahura, 12 for *Porites* in Tiahura; no estimable values exist for *Porites* out-transplants into Opunohu (n.d.).

5. Diversity is flat while membership turns over almost completely. Neither Shannon diversity nor Hill q1 responds to site, timeline or their interaction in either host (mixed models with colony as random effect, Type III tests: all $p \geq 0.27$; Tukey contrasts $p \geq 0.43$). Yet Baselga partitioning gives total Sørensen dissimilarity of 0.943 (*Pocillopora*) and 0.952 (*Porites*), of which **98.0% and 98.5% is pure species turnover** rather than nestedness — communities are replaced, not depleted.

6. *Endozoicomonas* loss marks the impacted bay in *Pocillopora*. The *Pocillopora* transplant contrast yields seven differentially abundant ASVs, of which **three are *Endozoicomonas*, all depleted in Opunohu** ($\log_2$FC −23.2 to −23.9, padj $\leq 2.6 \times 10^{-13}$), against enrichment of *Pseudoalteromonas* and a Xenococcaceae cyanobacterium. Given this genus's status as a core *Pocillopora* symbiont, it is the clearest candidate dysbiosis signal here. Other contrasts: 19 ASVs (*Porites* transplant, 15 enriched in Opunohu), 12 (*Pocillopora* heat-shock window), 173 (*Porites* heat-shock window, 138 depleted).

Limitations that bear on interpretation

1 **Dispersion confounds result 1.** Betadisper is significant for *Subtype* ($F = 10.74$, $p = 0.001$), so part of that result may be unequal within-group spread rather than centroid separation. Site dispersion is homogeneous ($F = 0.197$, $p = 0.653$), so results 2 and 4 are unaffected.

2 **The acute heat shock was not testable as specified.** In both heat-shock PERMANOVAs the *Treatment* term and interaction drop out, leaving site alone (*Pocillopora* $R^2 = 9.4\%$, $p = 0.077$, $n = 20$; *Porites* $p = 0.5$ on only **7 possible permutations**, $n = 8$). No heat-shock effect should be claimed from these models, and the "heat shock" differential-abundance tables are site contrasts *within* that window, not pre- versus post-heat comparisons.

3 **Differential abundance rests on one method.** LEfSe and MaAsLin2 returned zero significant features in all four contrasts, so the three-method consensus reduces to DESeq2 alone. Fold changes are unshrunken (no `lfcShrink`), which is why magnitudes reach $\log_2$FC ≈ 23; report direction, not magnitude.

4 **Sequencing depth is shallow.** Analysis depths of 2,066–3,072 reads per sample under-sample rare taxa.

5 **The reciprocal design is incomplete for *Porites*.** No tracking values are estimable for *Porites* fragments transplanted into Opunohu, so result 4 is a one-host statement about the impacted bay.

Next steps

Recode the heat term so it is estimable within colony blocks; apply `lfcShrink` before quoting any effect size; and either recover the Opunohu *Porites* out-transplants or declare the asymmetry explicitly in the manuscript.